

```
In [12]: # =====  
# Notebook setup  
# =====  
  
%load_ext autoreload  
%autoreload 2  
  
# Control figure size  
figsize=(15, 4)  
  
from util import util  
import pandas as pd  
import os  
  
# Load data  
data_folder = os.path.join('.', 'data', 'nab')  
file_name = os.path.join('realTraffic', 'occupancy_6005.csv')  
data, labels, windows = util.load_series(file_name, data_folder)
```

The autoreload extension is already loaded. To reload it, use:
%reload_ext autoreload

Missing Data in Time Series

Traffic Data, Again

Say we are contacted from a local transportation authority



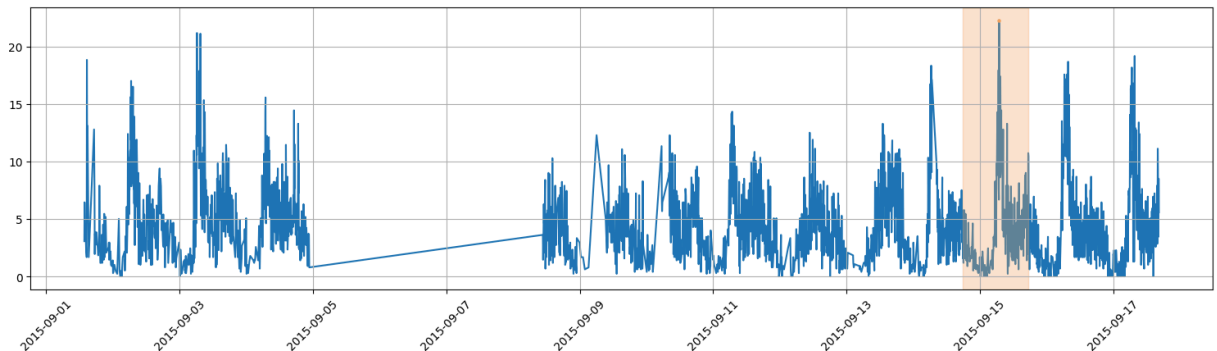
They want to improve their traffic monitoring system

Traffic Data, Again

They give us data from an occupancy sensor

Our data refers to real traffic in the Minnesota Twin Cities Area

```
In [4]: util.plot_series(data, labels, windows, figsize=figsize)
```

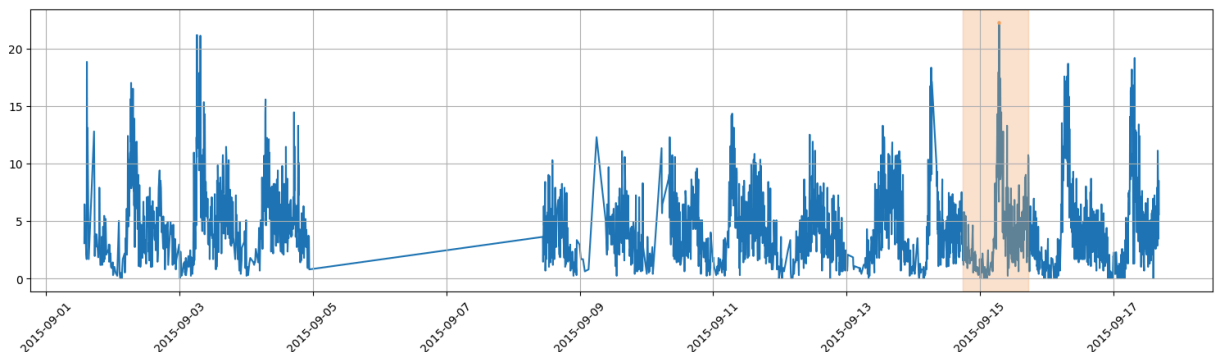


- They have pre-labeled an (easy) anomaly that they wish to detect
- ...But that is *not the most striking aspect* of this series

Traffic Data, Again

There is a *period*, and *straight lines in the plot*

```
In [5]: util.plot_series(data, labels, windows, figsize=figsize)
```

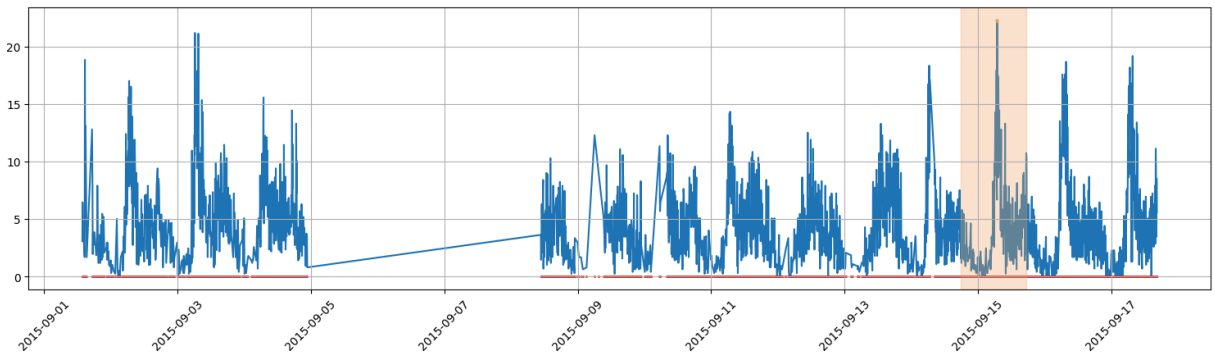


They are *artefacts*, due to *missing values* in the time series

Missing Values

We can make it clearer by explicitly plotting the sampling points

```
In [6]: util.plot_series(data, labels, windows, show_sampling_points=True, figsize=f
```



There is a large gap, plus scattered missing values here and there

Missing Values in Time Series

Missing values in real-world time series are *very common*

They arise for a variety of reasons:

- Malfunctioning sensors
- Network problems
- Lost data
- Sensor maintenance/installation/removal
- ...

...And can be very annoying to deal with

- They prevent the application of sliding windows
- They complicate the detection of periods
- ...

Preparing the Ground

Preparing the Ground

Before we can deal with missing values we need to tackle an issue

I.e. our main series has a *sparse index*

- ...Meaning that index values are non-contiguous
- ...And missing values are represented as gaps

If we want to fill the missing values...

- ...We need to decide *where* the missing values are

In other words, we need a *dense* (temporal) index

With a dense index:

- Missing values can be represented as NaN (Not a Number)
- ...And can be filled by replacing NaN with a meaningful value

Choosing a Sampling Frequency

First, we need to pick a frequency for the new index

We start by having a look at the typical sampling step in our series:

```
In [7]: data.head()
```

```
Out[7]:
```

	value
timestamp	
2015-09-01 13:45:00	3.06
2015-09-01 13:50:00	6.44
2015-09-01 13:55:00	5.17
2015-09-01 14:00:00	3.83
2015-09-01 14:05:00	4.50

- The interval between consecutive measurements seems to be 5 minute long
- ...But looking at just a few data points is not enough

Choosing a Sampling Frequency

It is much better to compute the distance between consecutive index values

```
In [8]: delta = data.index[1:] - data.index[:-1]
delta[:3]
```

```
Out[8]: TimedeltaIndex(['0 days 00:05:00', '0 days 00:05:00', '0 days 00:05:00'], dtype='timedelta64[ns]', name='timestamp', freq=None)
```

- The difference between two `datetime` objects is a `timedelta` object
- They are all parts of [the datetime module](#)

Then we can check the *value counts*

- This can be done with the `value_counts` method

The methods returns a series:

- The index contains values
- The series data are the corresponding counts

Choosing a Sampling Frequency

Let's have a look at our value counts

```
In [9]: vc = pd.Series(delta).value_counts()  
vc.iloc[:10]
```

```
Out[9]: timestamp  
0 days 00:05:00    1754  
0 days 00:10:00     340  
0 days 00:15:00     106  
0 days 00:20:00      37  
0 days 00:04:00      26  
0 days 00:25:00      22  
0 days 00:06:00      18  
0 days 00:30:00       9  
0 days 00:35:00       8  
0 days 00:11:00       7  
Name: count, dtype: int64
```

By far the most common value is 5 minutes

- Some values are not multiples of 5 minutes (e.g. 4, 6, 11 minutes)
- I.e. they are *out of alignment*

Resampling the Original Dataset

Therefore, first we need to *realign* the original index

This is also called *resampling* (or *binning*), and can be done in pandas with:

```
DatetimeIndex.resample(rule=None, ...)
```

- `rule` specifies the length of each individual interval (or "bin")

Resample is an iterator: we need to choose what to do with each bin

E.g. compute the mean, stdev, take the first value

```
In [10]: ddata = data.resample('5min').mean()  
ddata.head()
```

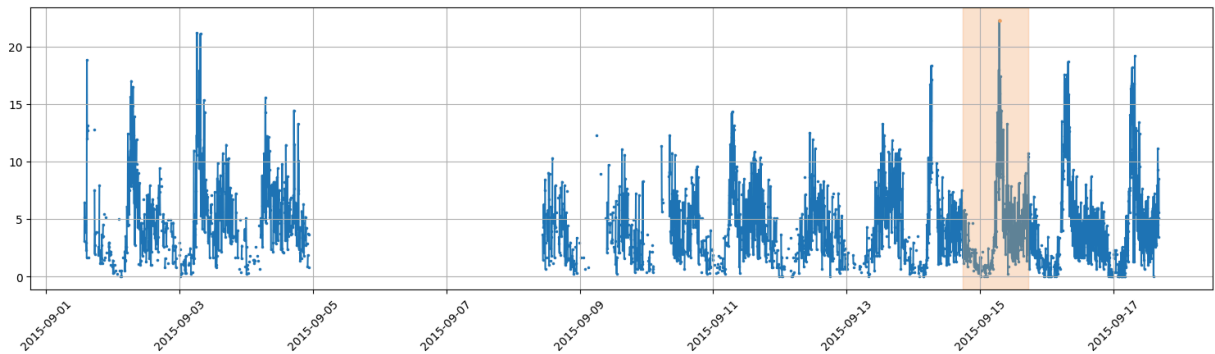
Out[10]:

	value
timestamp	
2015-09-01 13:45:00	3.06
2015-09-01 13:50:00	6.44
2015-09-01 13:55:00	5.17
2015-09-01 14:00:00	3.83
2015-09-01 14:05:00	4.50

Inspecting the Resampled Dataset

Now we can inspect this new "dense" series

```
In [11]: util.plot_series(ddata, labels, windows, show_markers=True, figsize=figsize)
```



- The artifacts have disappeared!
- ...And the true extent of our problem becomes apparent :-)